

Final: 12/19 Friday
8am-10am
@ BRWN 1154

STAT 516: Discrete Random Variables

Lecture 5: CDF/pmf, Medians, Basic moments

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Some simple distributions

- ▶ Roll a fair die and define $Y = 1$ if the sum of the two rolls is even
- ▶ Let $Y = 0$ if the sum of the two rolls is odd
- ▶ More formally, let A be any event in the sample space Ω
- ▶ Then,

$I_A = 1$ if A happens

$I_A = 0$ if A does not happen

- ▶ I_A is an **indicator** of A
- ▶ It is also a **Bernoulli random variable** with the parameter $p = P(A)$

Example

- ▶ X = number of **aces** in the first player's hand in a bridge game; X can be any of $x = 0, 1, 2, 3, 4$.
- ▶ The other $13 - x$ cards this player's hand must be non-ace cards Hypergeometric Distribution

- ▶ The pmf of X is

• $N = 52$ cards

• $D =$ Type Total of x (4 aces)
desired

• $n =$

• $x = \dots$

$$P(X = x) = \frac{\overset{\substack{\text{possible aces} \\ \text{only 4 total} \\ \uparrow \\ \text{4 choose } x}}{\binom{4}{x}} \overset{\substack{\text{not aces cards } (52-4=48) \\ \uparrow}}{\binom{48}{13-x}}}{\underset{\substack{\text{Total 52 cards,} \\ \text{we have total of 13 in hand}}}{\binom{52}{13}}} \quad \text{48 choose } 13-x$$

for any $x = 0, 1, 2, 3, 4$

Hypergeometric distribution

- ▶ This is an example of the **hypergeometric distribution**: 3 parameters: (N, D, n)
 1. among N total objects some D are of type I
 2. the other $N - D$ are of type II
 3. n are selected without replacement from the N objects
- ▶ The pmf of the number of type I objects among the n selected is

$X = \left\{ \begin{array}{l} \text{Number of Type I objects} \\ \text{among the } n \text{ selected} \end{array} \right\}$

$$P(X = x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}$$

Type I ← opposite Type II

where $n + D - N \leq x \leq D$

- $x \leq D$: "D choose x"
- $n - x \leq N - D$, meaning:
 $n + D - N \leq x$

Function of a random variable

- ▶ Let X be a discrete random variable and $Y = g(X)$ a real-valued function of X . Then, $P(Y = y) = \sum_{x: \overset{\text{stands for}}{g(x)} = y} \overset{\text{stands for}}{p(x)}$.
- ▶ Example: let $X \sim p(x)$ where $p(x) = \frac{c}{1+x^2}$ for some $c > 0$
- ▶ Here, $x = 0, \pm 1, \pm 2, \pm 3$.
- ▶ What is the distribution of $Y = g(X) = X^3$ and $Z = h(X) = \sin\left(\frac{\pi}{2}X\right)$?

Function of a random variable

$$\frac{1}{1+x^2} = \frac{1}{10} + \frac{1}{5} + \frac{1}{2} + \left(\frac{1}{2} + \frac{1}{5} + \frac{1}{10}\right) = \frac{13}{5}$$

$$\sum_{-3}^3 \left(\frac{c}{1+x^2} \right) = 1 \Rightarrow c \cdot \frac{13}{5} = 1 \Rightarrow c = \frac{5}{13}$$

$$p(x) = \begin{cases} \frac{5}{13(1+x^2)}, & x=0, \pm 1, \pm 2, \pm 3 \\ 0 & \text{elsewhere} \end{cases}$$

- First, by direct computation, because $\sum_x p(x) = 1$ we find that $c = \frac{5}{13}$
- Possible values of Y are $0, \pm 1, \pm 8, \pm 27$
- In general,

$$\sqrt[3]{y} = \sqrt[3]{x^3}$$

$$y^{1/3} = x$$

$$P(Y = y) = P(X = y^{1/3}) = \frac{c}{1 + y^{2/3}} = \frac{5}{13(1+x^2)} = \frac{5}{13(1+(y^{1/3})^2)}$$

with $c = \frac{5}{13}$

$$p(Y=y) = p(X=y^{1/3}) = \frac{5}{13(1+y^{2/3})} \text{ for any } y = 0, \pm 1, \pm 8, \pm 27$$

Independence of random variables

- ▶ Let $X_1, X_2, \dots, X_k$ be $k \geq 2$ discrete random variables on the same Ω
- ▶ $X_1, \dots, X_k$ are independent if

$$\begin{aligned} \text{Joint Probability } P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) \\ = P(X_1 = x_1)P(X_2 = x_2) \cdots P(X_k = x_k) \end{aligned}$$

Product of the simple probability
for all $x_1, x_2, \dots, x_k$

Independence of functions of random variables

► Theorem:

1. Let $X_1, X_2, \dots, X_k$ be $k \geq 2$ discrete random variables that are independent
2. Let $U = f(X_1, X_2, \dots, X_i)$ be some function of $X_1, X_2, \dots, X_i$ and $V = g(X_{i+1}, \dots, X_k)$ be some function of $X_{i+1}, \dots, X_k$
3. Then U and V are independent

A new (and useful!) notation

- ▶ Let $X_1, \dots, X_k$ be independent random variables with the same cdf F
- ▶ Then we say that $X_1, \dots, X_k$ are iid (IID) and write
 $X_1, X_2, \dots, X_k \stackrel{iid}{\sim} F$
*independent
identically distributed*

Expected value of a Discrete Random Variable

- ▶ Let X be a discrete random variable
- ▶ The **expected value** of X exists if $\sum |x_i|p(x_i) < \infty$ and is then defined as

$$\mu = E(X) = \sum_i x_i p(x_i)$$

- ▶ μ is also known as the **expectation** or the **mean** of X

Change of Variable Formula for discrete random variables

- If Ω is finite or countably infinite and X is a discrete random variable with expectation μ ,

$$\mu = \sum_{\substack{x \text{ over all possible} \\ \text{values of } X}} xp(x) = \sum_{\omega} X(\omega)P(\omega)$$

Example:

$$X = \begin{cases} 1, 1/2 \\ 0, 1/2 \end{cases}$$
$$\mathbb{E}(X) = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2}$$

where $P(\omega)$ is the probability of the sample point ω

Expectation of functions of several random variables

- ▶ Often, we want to find an expectation of e.g. a sum $X + Y$ or a product XY
- ▶ Let $X_1, X_2, \dots, X_n$ be n discrete random variables defined on a common sample space Ω
- ▶ The expectation of a function $g(X_1, X_2, \dots, X_n)$ exists if $\sum_{\omega} |g(X_1(\omega), \dots, X_n(\omega))| P(\omega) < \infty$ in which case

$$E[g(X_1, X_2, \dots, X_n)] = \sum_{\omega} g(X_1(\omega), X_2(\omega), \dots, X_n(\omega)) P(\omega)$$

Absolute value
 $g(x)(y)$

Example

- ▶ X is the number of ones and Y the number of sixes in two rolls of a fair die; consider $g(X, Y) = XY$
- ▶ The sample space is $\Omega = \{(1, 1), (1, 2), (1, 3), \dots, (6, 4), (6, 5), (6, 6)\}$
- ▶ $P(\omega) = \frac{1}{36}$ for each $\omega \in \Omega$; the values of 36 sample points are $X(\omega_1)Y(\omega_1) = 2 \times 0 = 0, X(\omega_2)Y(\omega_2) = 1 \times 0 = 0$ etc.
- ▶ $(1, 1) = 2 \text{ 1's for } X = 2 \times 0 = 0$ $(1, 2)$

$$E(XY) = 0 \times \underbrace{\frac{1}{36}}_{w_1} + 0 \times \underbrace{\frac{1}{36}}_{w_2} + \dots + 0 \times \frac{1}{36} = \frac{1}{18}$$

$$\frac{X(\omega_3)Y(\omega_3) = 1 \times 0 = 0}{(1, 3)}$$

$$\frac{\circ X(\omega_q)Y(\omega_q) = 1 \times 0 = 0}{(1,4)}$$

$$\begin{aligned}(1, 6) &= 1 \\ (6, 1) &= 1 \\ \text{Nonzero values only } (1, 6), (6, 1) \\ \frac{2}{36} &= \frac{1}{18}\end{aligned}$$